Capstone Project Bloom Predictions

Amie Bui

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—————————Washington, DC: Build Model——————————

```
dc_weather <- read.csv("~/STAT490 Capstone/USW00013743.csv")
```

Clean Data set

```
dc_weather_clean <- dc_weather %>%
  mutate(
    tmax_c = (tmax - 32) * 5/9,
    tmin_c = (tmin - 32) * 5/9,
    Date = as.Date(Date),
    year = year(Date),
    TMEAN = (tmax_c + tmin_c) / 2
  ) %>%
  filter(!is.na(TMEAN))

head(dc_weather_clean)
```

```
##     X       Date tmax tmin prcp    tmax_c    tmin_c year    TMEAN
## 1 366 1872-01-01   52   34 0.04 11.111111  1.111111 1872 6.111111
## 2 367 1872-01-02   41   29 0.00  5.000000 -1.666667 1872 1.666667
## 3 368 1872-01-03   44   32 0.03  6.666667  0.000000 1872 3.333333
## 4 369 1872-01-04   43   37 0.00  6.111111  2.777778 1872 4.444444
## 5 370 1872-01-05   44   32 0.00  6.666667  0.000000 1872 3.333333
## 6 371 1872-01-06   47   29 0.00  8.333333 -1.666667 1872 3.333333
```

Bloom Dates

```
bloom_url <- "https://raw.githubusercontent.com/GMU-CherryBlossomCompetition/peak-bloom-prediction/main/
blooms <- read.csv(bloom_url)
```

Calculate Chill Threshold

```
chill_threshold <- 0


dc_weather_clean <- dc_weather_clean %>%
  mutate(
    Date = as.Date(Date),
    month_day = format(Date, "%m-%d"),
    #shift so chill corresponds with correct year
    bloom_year = year(Date) + if_else(month(Date) >= 11, 1L, 0L),
  )



chill_data <- dc_weather_clean %>%
  filter(month_day >= "11-01" | month_day <= "01-31") %>%
  group_by(bloom_year) %>%
  summarise(chill_days = sum(TMEAN < chill_threshold))
```

Compute Growing Degree Days

i. If temp is greater than 45 degrees fahrenheit, 7.22 degrees Celsius, count it is counted as "warmth"

```
gdd_data <- dc_weather_clean %>%
  filter(month_day >= "02-01" & month_day <= "04-30") %>%
  mutate(gdd = pmax(TMEAN - 7.22, 0)) %>%
  group_by(bloom_year) %>%
  summarise(total_heat = sum(gdd))
```

Merge data

```
merged_data_dc <- chill_data %>%
  left_join(gdd_data, by = "bloom_year") %>%
  left_join(blooms, by = c("bloom_year" = "year"))%>%
  select(-location, -lat, -long, -alt) %>%
  drop_na(bloom_doy, chill_days, total_heat)
```

Split into test and training data

```
merged_data_dc <- merged_data_dc %>% arrange(bloom_year)

n <- nrow(merged_data_dc)
split_index <- floor(0.8 * n)

train_data_dc <- merged_data_dc[1:split_index, ]
test_data_dc  <- merged_data_dc[(split_index + 1):n, ]
```

Fit Model DC

```r
dc_model <- lm(bloom_doy ~ chill_days + total_heat + total_heat * chill_days,
             data = train_data_dc)
```

Testing on Held Out Test Set

```r
test_data_dc$predicted_bloom <- predict(dc_model, newdata = test_data_dc)
MAE <- mean(abs(test_data_dc$bloom_doy - test_data_dc$predicted_bloom))
MAE
```

```
## [1] 3.298717
```

Baseline MAE

```r
baseline_pred_dc <- mean(train_data_dc$bloom_doy)

test_data_dc$baseline_pred <- baseline_pred_dc

baseline_MAE_dc <- mean(abs(test_data_dc$bloom_doy - test_data_dc$baseline_pred))
baseline_MAE_dc
```

```
## [1] 5.71
```

————————-Kyoto, Japan: Build Model————————–

Get Weather Data for Kyoto Japan

```r
kyoto_weather <- get_power(
  community = "AG",
  temporal_api = "daily",
  lonlat = c(135.7681, 35.0116), #Location of Kyoto
  pars = c("T2M_MAX", "T2M_MIN"),
  dates = c("1981-01-01", "2023-12-31")
)

kyoto_weather <- kyoto_weather %>%
  mutate(TMEAN = (T2M_MAX + T2M_MIN) / 2)
```

Bloom Dates Kyoto Japan

```r
kyoto <- read.csv("~/STAT490 Capstone/kyoto.csv")
```

Calculate Chill Threshold

```
chill_threshold <- 0


kyoto_weather <- kyoto_weather %>%
  mutate(
    #Shift chill over so it is associated with the correct bloom year
    bloom_year = if_else(MM >= 11, YEAR + 1L, YEAR)
  )


chill_data_japan <- kyoto_weather %>%
  filter(MM >= 11| MM <= 1) %>%
  group_by(bloom_year) %>%
  summarise(chill_days = sum(TMEAN < chill_threshold))
```

Compute Growing Degree Days

i. If temp is greater than 7.22 degrees Celsius, it is counted as "warmth"

```
gdd_data_japan <- kyoto_weather %>%
  filter(MM >= 2 & MM <=  4) %>%
  mutate(gdd = pmax(TMEAN - 7.222, 0)) %>%
  group_by(bloom_year) %>%
  summarise(total_heat = sum(gdd))
```

Merge data

```
merged_data_japan <- chill_data_japan %>%
  left_join(gdd_data_japan, by = "bloom_year") %>%
  left_join(kyoto, by = c("bloom_year" = "year"))%>%
  select(-location, -lat, -long, -alt) %>%
  drop_na(bloom_doy, chill_days, total_heat)
```

Split into test and training data

```
merged_data_japan <- merged_data_japan %>% arrange(bloom_year)

n <- nrow(merged_data_japan)
split_index <- floor(0.8 * n)

train_data_japan <- merged_data_japan[1:split_index, ]
test_data_japan  <- merged_data_japan[(split_index + 1):n, ]
```

Fit Model Japan

```
japan_model<- lm(bloom_doy ~ chill_days + total_heat + total_heat * chill_days,
             data = train_data_japan)
```

Testing on Held Out Test Set with Model

```
test_data_japan$predicted_bloom <- predict(japan_model, newdata = test_data_japan)
MAE <- mean(abs(test_data_japan$bloom_doy - test_data_japan$predicted_bloom))
MAE
```

```
## [1] 3.017026
```

Baseline MAE

```
baseline_pred_japan <- mean(train_data_japan$bloom_doy)

test_data_japan$baseline_pred <- baseline_pred_japan

baseline_MAE_japan <- mean(abs(test_data_japan$bloom_doy - test_data_japan$baseline_pred))
baseline_MAE_japan
```

```
## [1] 5.977124
```

—————————Switzerland: Build Model—————————-

Get Weather Data for Liestal, Switzerland

```
switzerland_weather <- get_power(
  community = "AG",
  temporal_api = "daily",
  lonlat = c(7.735, 47.484), #Location of Liestal
  pars = c("T2M_MAX", "T2M_MIN"),
  dates = c("1981-01-01", "2023-12-31")
)

switzerland_weather <- switzerland_weather %>%
  mutate(TMEAN = (T2M_MAX + T2M_MIN) / 2)
```

Bloom Dates Liestal, Switzerland

```
liestal <- read.csv("~/STAT490 Capstone/liestal.csv")
```

Calculate Chill Threshold

```
chill_threshold <- 0


switzerland_weather <- switzerland_weather %>%
  mutate(
    #Shift chill over so it is associated with the correct bloom year
    bloom_year = if_else(MM >= 11, YEAR + 1L, YEAR)
  )


chill_data_switzerland <- switzerland_weather %>%
  filter(MM >= 11| MM <= 1) %>%
  group_by(bloom_year) %>%
  summarise(chill_days = sum(TMEAN < chill_threshold))
```

Compute Growing Degree Days

i. If temp is greater than 45 degrees fahrenheit (7.222 degrees Celsius), it is counted as "warmth"

```
gdd_data_switzerland <- switzerland_weather %>%
  filter(MM >= 2 & MM <=  4) %>%
  mutate(gdd = pmax(TMEAN - 7.222, 0)) %>%
  group_by(bloom_year) %>%
  summarise(total_heat = sum(gdd))
```

Merge data

```
merged_data_switzerland <- chill_data_switzerland %>%
  left_join(gdd_data_switzerland, by = "bloom_year") %>%
  left_join(liestal, by = c("bloom_year" = "year"))%>%
  select(-location, -lat, -long, -alt) %>%
  drop_na(bloom_doy, chill_days, total_heat)
```

Split into test and training data

```
merged_data_switzerland <- merged_data_switzerland %>% arrange(bloom_year)

n <- nrow(merged_data_switzerland)
split_index <- floor(0.8 * n)

train_data_switzerland <- merged_data_switzerland[1:split_index, ]
test_data_switzerland  <- merged_data_switzerland[(split_index + 1):n, ]
```

Fit Model Switzerland

```r
liestal_model<- lm(bloom_doy ~ chill_days + total_heat + total_heat * chill_days,
             data = train_data_switzerland)
```

Testing on Held Out Test Set

```r
test_data_switzerland$predicted_bloom <- predict(liestal_model, newdata = test_data_switzerland)
MAE <- mean(abs(test_data_switzerland$bloom_doy - test_data_switzerland$predicted_bloom))
MAE
```

```
## [1] 8.47237
```

Baseline MAE

```r
baseline_pred_switzerland <- mean(train_data_switzerland$bloom_doy)

test_data_switzerland$baseline_pred <- baseline_pred_switzerland

baseline_MAE_switzerland <- mean(abs(test_data_switzerland$bloom_doy - test_data_switzerland$baseline_p
baseline_MAE_switzerland
```

```
## [1] 8.493464
```

———————Vancouver: Clean Dataset———————-

Get Weather Data for Vancouver

```r
vancouver_weather <- get_power(
  community = "AG",
  temporal_api = "daily",
  lonlat = c(-123.1636, 49.2237), #Location of Vancouver
  pars = c("T2M_MAX", "T2M_MIN"),
  dates = c("1981-01-01", "2023-12-31")
)

vancouver_weather <- vancouver_weather %>%
  mutate(TMEAN = (T2M_MAX + T2M_MIN) / 2)
```

Bloom Data for Vancouver

```r
vancouver <- read.csv("~/STAT490 Capstone/vancouver.csv")
```

Calculate Chill Threshold

```
chill_threshold <- 0


vancouver_weather <- vancouver_weather %>%
  mutate(
    #Shift chill over so it is associated with the correct bloom year
    bloom_year = if_else(MM >= 11, YEAR + 1L, YEAR)
  )


chill_data_vancouver <- vancouver_weather %>%
  filter(MM >= 11| MM <= 1) %>%
  group_by(bloom_year) %>%
  summarise(chill_days = sum(TMEAN < chill_threshold))
```

Compute Growing Degree Days

i. If temp is greater than 45 degrees fahrenheit (7.222 degrees Celsius), it is counted as "warmth"

```
gdd_data_vancouver <- vancouver_weather %>%
  filter(MM >= 2 & MM <=  4) %>%
  mutate(gdd = pmax(TMEAN - 7.222, 0)) %>%
  group_by(bloom_year) %>%
  summarise(total_heat = sum(gdd))
```

Merge data: Used Only for External Validation

```
merged_data_vancouver <- chill_data_vancouver %>%
  left_join(gdd_data_vancouver, by = "bloom_year") %>%
  left_join(vancouver, by = c("bloom_year" = "year"))%>%
  select(-location, -lat, -long, -alt) %>%
  drop_na(bloom_doy, chill_days, total_heat)
```

——————————Build Pooled Model——————————

Used this Pooled Model To Predict for Vancouver and NYC

Merge All Countries Train Data Set

```
train_data_all <- bind_rows(train_data_dc, train_data_japan,
                                    train_data_switzerland)
```

Fit Model

```
pooledModel <- lm(bloom_doy ~ chill_days + total_heat + chill_days*total_heat,
             data = train_data_all)
```

Testing on Held Out Test Set

Washington

```
test_data_dc$predicted_bloom <- predict(pooledModel, newdata = test_data_dc)
MAE <- mean(abs(test_data_dc$bloom_doy - test_data_dc$predicted_bloom))
MAE
```

```
## [1] 3.874648
```

Vancouver

```
merged_data_vancouver$predicted_bloom <- predict(pooledModel, newdata = merged_data_vancouver)
MAE <- mean(abs(merged_data_vancouver$bloom_doy - merged_data_vancouver$predicted_bloom))
MAE
```

```
## [1] 9.603864
```

——————————Washington Predictions 2026——————————

Import Weather Forecast Data Sets and Filter

```
accuweather_2026 <- read.csv("~/STAT490 Capstone/accuweather_forecast_2026.csv")
EOY_Weather_2025 <- read.csv("~/STAT490 Capstone/EOY_Weather_2025.csv")

weather_pred_DC_2026 <- accuweather_2026 %>%
  filter(location == "washington") %>%
  mutate(TMEAN = (tmax + tmin)/2)

weather_DC_2025 <- EOY_Weather_2025 %>%
  filter(location == "washington") %>%
  mutate(TMEAN = (TMAX_C + TMIN_C)/2)
```

Calculate Chill

```
chill_threshold <- 0


chill_washington_2026 <-sum(weather_DC_2025$TMEAN < chill_threshold) +
  sum(weather_pred_DC_2026$TMEAN < chill_threshold &
        month(weather_pred_DC_2026$date) <= 1)
```

Calculate Growing Days

```
gdd_pred_US_2026 <- weather_pred_DC_2026 %>%
  filter(month(weather_pred_DC_2026$date) >= 2 & month(weather_pred_DC_2026$date) <= 4) %>%
  mutate(gdd = pmax(TMEAN - 7.222, 0)) %>%
  summarise(total_heat = sum(gdd))
```

Prediction

```
newdata_washington  <- data.frame(
  chill_days = chill_washington_2026,
  total_heat = gdd_pred_US_2026$total_heat
)

pred_doy_washington = predict(dc_model, newdata = newdata_washington)
as.Date(pred_doy_washington - 1, origin = "2026-01-01")
```

```
##            1
## "2026-04-05"
```

Japan Prediction 2026

Import Weather Forecast Data Sets and Filter

```
accuweather_2026 <- read.csv("~/STAT490 Capstone/accuweather_forecast_2026.csv")
EOY_Weather_2025 <- read.csv("~/STAT490 Capstone/EOY_Weather_2025.csv")

weather_pred_japan_2026 <- accuweather_2026 %>%
  filter(location == "kyoto") %>%
  mutate(TMEAN = (tmax + tmin)/2)

weather_japan_2025 <- EOY_Weather_2025 %>%
  filter(location == "kyoto") %>%
  mutate(TMEAN = (TMAX_C + TMIN_C)/2)
```

Calculate Chill

```r
chill_threshold <- 0


chill_japan_2026 <-sum(weather_japan_2025$TMEAN < chill_threshold) +
  sum(weather_pred_japan_2026$TMEAN < chill_threshold &
        month(weather_pred_japan_2026$date) <= 1)
```

Calculate Growing Days

```r
gdd_pred_japan_2026 <- weather_pred_japan_2026 %>%
  filter(month(weather_pred_japan_2026$date) >= 2 & month(weather_pred_japan_2026$date) <= 4) %>%
  mutate(gdd = pmax(TMEAN - 7.222, 0)) %>%
  summarise(total_heat = sum(gdd))
```

Prediction

```r
newdata_japan <- data.frame(
  chill_days = chill_japan_2026,
  total_heat = gdd_pred_japan_2026$total_heat
)

pred_doy_japan = predict(japan_model, newdata = newdata_japan)
as.Date(pred_doy_japan - 1, origin = "2026-01-01")
```

```
##            1
## "2026-04-01"
```

——————————Vancouver Prediction——————————

Import Weather Forecast Data Sets and Filter

```r
accuweather_2026 <- read.csv("~/STAT490 Capstone/accuweather_forecast_2026.csv")
EOY_Weather_2025 <- read.csv("~/STAT490 Capstone/EOY_Weather_2025.csv")

weather_pred_vancouver_2026 <- accuweather_2026 %>%
  filter(location == "vancouver") %>%
  mutate(TMEAN = (tmax + tmin)/2)

weather_vancouver_2025 <- EOY_Weather_2025 %>%
  filter(location == "vancouver") %>%
  mutate(TMEAN = (TMAX_C + TMIN_C)/2)
```

Calculate Chill

```
chill_threshold <- 0


chill_vancouver_2026 <-sum(weather_vancouver_2025$TMEAN < chill_threshold) +
  sum(weather_pred_vancouver_2026$TMEAN < chill_threshold &
        month(weather_pred_vancouver_2026$date) <= 1)
```

Calculate Growing Days

```
gdd_pred_vancouver_2026 <- weather_pred_vancouver_2026 %>%
  filter(month(weather_pred_vancouver_2026$date) >= 2 & month(weather_pred_vancouver_2026$date) <= 4) %
  mutate(gdd = pmax(TMEAN - 7.222, 0)) %>%
  summarise(total_heat = sum(gdd))
```

Prediction

```
newdata_vancouver <- data.frame(
  chill_days = chill_vancouver_2026,
  total_heat = gdd_pred_vancouver_2026$total_heat
)

pred_doy_vancouver = predict(pooledModel, newdata = newdata_vancouver)
as.Date(pred_doy_vancouver - 1, origin = "2026-01-01")
```

```
##            1
## "2026-04-08"
```

———————————-Prediction Switzerland 2026———————————-

Import Weather Forecast Data Sets and Filter

```
accuweather_2026 <- read.csv("~/STAT490 Capstone/accuweather_forecast_2026.csv")
EOY_Weather_2025 <- read.csv("~/STAT490 Capstone/EOY_Weather_2025.csv")

weather_pred_liestal_2026 <- accuweather_2026 %>%
  filter(location == "liestal") %>%
  mutate(TMEAN = (tmax + tmin)/2)

weather_liestal_2025 <- EOY_Weather_2025 %>%
  filter(location == "liestal") %>%
  mutate(TMEAN = (TMAX_C + TMIN_C)/2)
```

Calculate Chill

```r
chill_threshold <- 0


chill_liestal_2026 <-sum(weather_liestal_2025$TMEAN < chill_threshold) +
  sum(weather_pred_liestal_2026$TMEAN < chill_threshold &
        month(weather_pred_liestal_2026$date) <= 1)
```

Calculate Growing Days

```r
gdd_pred_liestal_2026 <- weather_pred_liestal_2026 %>%
  filter(month(weather_pred_liestal_2026$date) >= 2 & month(weather_pred_liestal_2026$date) <= 4) %>%
  mutate(gdd = pmax(TMEAN - 7.222, 0)) %>%
  summarise(total_heat = sum(gdd))
```

Prediction

```r
newdata_liestal <- data.frame(
  chill_days = chill_liestal_2026,
  total_heat = gdd_pred_liestal_2026$total_heat
)

pred_doy_liestal = predict(liestal_model, newdata = newdata_liestal)
as.Date(pred_doy_liestal - 1, origin = "2026-01-01")
```

```
##            1
## "2026-04-02"
```

———————Prediction New York City——————————

Import Weather Forecast Data Sets and Filter

```r
accuweather_2026 <- read.csv("~/STAT490 Capstone/accuweather_forecast_2026.csv")
EOY_Weather_2025 <- read.csv("~/STAT490 Capstone/EOY_Weather_2025.csv")

weather_pred_nyc_2026 <- accuweather_2026 %>%
  filter(location == "newyork") %>%
  mutate(TMEAN = (tmax + tmin)/2)

weather_newyork_2025 <- EOY_Weather_2025 %>%
  filter(location == "newyork") %>%
  mutate(TMEAN = (TMAX_C + TMIN_C)/2)
```

Calculate Chill

```r
chill_threshold <- 0


chill_nyc_2026 <-sum(weather_newyork_2025$TMEAN < chill_threshold) +
  sum(weather_pred_nyc_2026$TMEAN < chill_threshold &
        month(weather_pred_nyc_2026$date) <= 1)
```

Calculate Growing Days

```r
gdd_pred_nyc_2026 <- weather_pred_nyc_2026 %>%
  filter(month(weather_pred_nyc_2026$date) >= 2 & month(weather_pred_nyc_2026$date) <= 4) %>%
  mutate(gdd = pmax(TMEAN - 7.222, 0)) %>%
  summarise(total_heat = sum(gdd))
```

Prediction

```r
newdata_nyc <- data.frame(
  chill_days = chill_nyc_2026,
  total_heat = gdd_pred_nyc_2026$total_heat
)

pred_doy_nyc = predict(pooledModel, newdata = newdata_nyc)
as.Date(pred_doy_nyc - 1, origin = "2026-01-01")
```

```
##            1
## "2026-04-06"
```